

AYUSH KUMAR

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PROFESSIONAL SUMMARY

Designed, built, and deployed a production-grade GenAI application with RAG pipelines, tool-using agents, multi-step orchestration, and structured API microservices. MSc Advanced Computer Science student with hands-on experience in LLM application patterns including prompt engineering, function calling, guardrails, evaluation, and constrained decoding. Strong Python and SQL proficiency with production deployment experience using FastAPI, Docker, Git, and CI/CD practices. Published at IEEE ICBDMML 2024.

EDUCATION

University of Sheffield MSc Advanced Computer Science	Sheffield, UK September 2025 - Present
Chandigarh University Bachelor of Engineering, Computer Science	Chandigarh, India August 2021 - June 2025

TECHNICAL SKILLS

GenAI and Agentic AI: RAG Pipelines, Tool-Using Agents, Multi-Step Orchestration, Prompt Engineering, Function Calling, Tool Routing, Guardrails, Constrained Decoding, Structured Output Parsing, LLM Evaluation, Embeddings, Vector Search

Languages: Python (Advanced), SQL, C++, C#, JavaScript

APIs and Microservices: FastAPI, REST API Design, gRPC Fundamentals, JSON Schema Validation, Service Integration, Flask, Next.js, React.js

ML and MLOps: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, ONNX Runtime, Model Versioning, Experiment Tracking, LoRA Fine-Tuning Fundamentals

Infrastructure: Docker, Git, GitHub, Linux, CI/CD, PostgreSQL, MongoDB, Automated Testing, Logging, Monitoring, Observability Fundamentals

PROJECTS

Production GenAI Agent with RAG Pipeline (Python, FastAPI)	2025
- Designed and deployed production-grade agentic AI system: RAG-based retrieval, tool-using agent with function calling, multi-step orchestration with planning, routing, retries, fallbacks, and state management across pipeline stages - Built prompt engineering assets including structured templates, prompt routing logic, and output guardrails with JSON schema validation, format constraints, and automated evaluation to ensure quality across runs - Engineered production microservice with FastAPI backend exposing REST API endpoints, structured request/response schemas, error handling, and logging, serving real users with sub-second latency - Implemented end-to-end ML lifecycle from data exploration and pipeline design through deployment and monitoring, with clean testable code, version control via Git, and CI/CD practices	
ML-Powered Decision Service with Cross-Platform Deployment (Python, PyTorch, C#)	2025 - Present
- Built production service combining PyTorch-trained classification model deployed via ONNX Runtime with algorithmic decision logic, managing model serialization, versioning, and cross-platform inference - Implemented comprehensive testing (unit and integration), performance benchmarking across configurations, and structured logging enabling observability of model predictions and system behaviour	
Automated ML Pipeline with Evaluation Framework (Python, TensorFlow)	2024
Developed end-to-end ML pipeline covering data processing, model training, hyperparameter optimization, and automated evaluation on NIH datasets (89.15% accuracy), with systematic error analysis and documented experimental methodology	
Real-Time Classification Service (Python, OpenCV)	2023
Built real-time classification service processing 1,000+ inputs from hardware sensor at 90.68% accuracy, with data-driven debugging, failure mode identification, and reliability improvements	

PUBLICATION

Tulsyan, R., Shukla, P., Singh, T., and Kumar, A. (2024). The Impact of JavaScript Frameworks on Website Performance and User Experience. 2024 IEEE International Conference on Big Data and Machine Learning (ICBDML), 299-305.